

# EC centrifugal fan

backward-curved, single-intake

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## Nominal data

|                          |                   |            |
|--------------------------|-------------------|------------|
| Type                     | R3G190-RC05-49    |            |
| Motor                    | M3G055-BI         |            |
| Phase                    |                   | 1~         |
| Nominal voltage          | VAC               | 230        |
| Nominal voltage range    | VAC               | 200 .. 240 |
| Frequency                | Hz                | 50/60      |
| Method of obtaining data |                   | ml         |
| Speed (rpm)              | min <sup>-1</sup> | 3200       |
| Power consumption        | W                 | 83         |
| Current draw             | A                 | 0.75       |
| Min. ambient temperature | °C                | -25        |
| Max. ambient temperature | °C                | 60         |

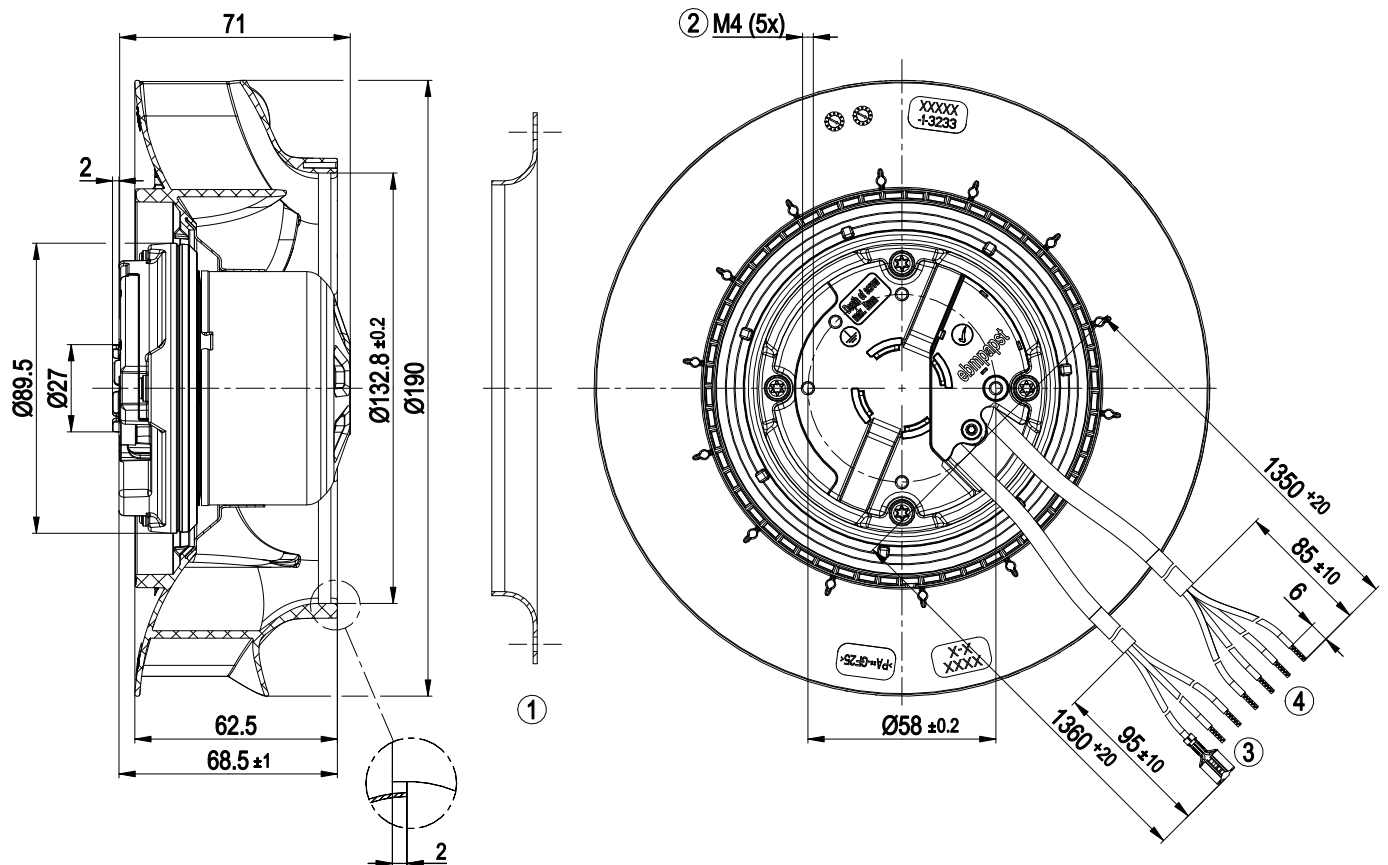
ml = Max. load · me = Max. efficiency · fa = Free air · cs = Customer specification · ce = Customer equipment  
Subject to change



## Technical description

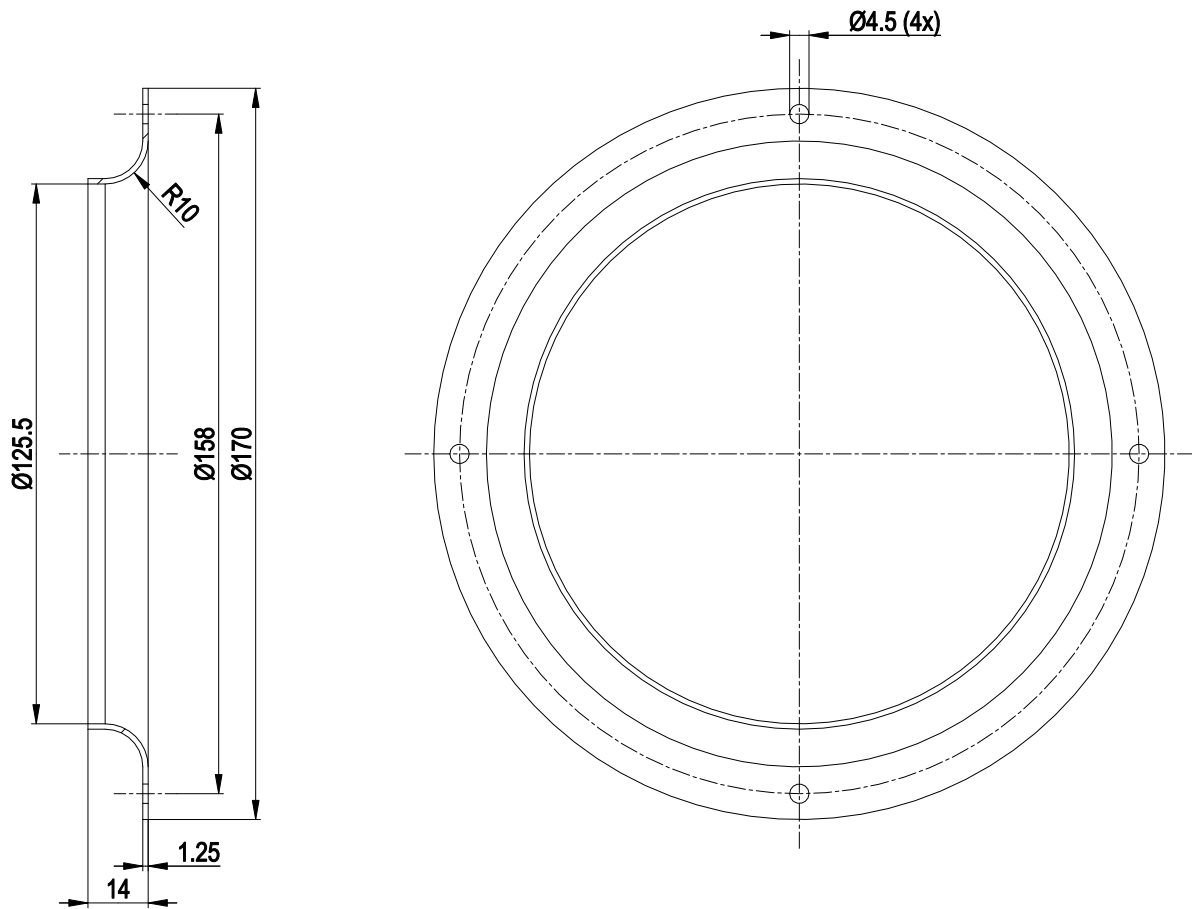
|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weight                                                                     | 1.2 kg                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Size                                                                       | 190 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Motor size                                                                 | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Rotor surface                                                              | Thick-film passivated                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Impeller material                                                          | PP plastic                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Number of blades                                                           | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Direction of rotation                                                      | Clockwise, viewed toward rotor                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Degree of protection                                                       | IP54                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Insulation class                                                           | "B"                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Moisture (F) / Environmental (H) protection class                          | H1+                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Max. permitted ambient temp. for motor (transport/storage)                 | + 80 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Min. permitted ambient temp. for motor (transport/storage)                 | - 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Installation position                                                      | Any                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Condensation drainage holes                                                | None, open rotor                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Mode                                                                       | S1                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Motor bearing                                                              | Ball bearing; (sealed)                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Technical features                                                         | <ul style="list-style-type: none"> <li>- Output 10 VDC, max. 1.1 mA</li> <li>- Tach output</li> <li>- Power limiter</li> <li>- Motor current limitation</li> <li>- Soft start</li> <li>- Control input 0-10 VDC / PWM</li> <li>- Control interface with SELV potential safely disconnected from the mains</li> <li>- Overvoltage detection</li> <li>- Thermal overload protection for electronics/motor</li> <li>- Line undervoltage detection</li> </ul> |
| EMC immunity to interference                                               | According to EN 61000-6-2 (industrial environment)                                                                                                                                                                                                                                                                                                                                                                                                        |
| EMC circuit feedback                                                       | According to EN 61000-3-2/3                                                                                                                                                                                                                                                                                                                                                                                                                               |
| EMC interference emission                                                  | According to EN 61000-6-3 (household environment)                                                                                                                                                                                                                                                                                                                                                                                                         |
| Touch current according to IEC 60990 (measuring circuit Fig. 4, TN system) | <= 3.5 mA                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Motor protection                                                           | Electronic motor protection                                                                                                                                                                                                                                                                                                                                                                                                                               |
| With cable                                                                 | Variable                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Protection class                                                           | I (with customer connection of protective earth)                                                                                                                                                                                                                                                                                                                                                                                                          |
| Conformity with standards                                                  | EN 60034-1; EN 60204-1; EN 60335-1; CE                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Product drawing



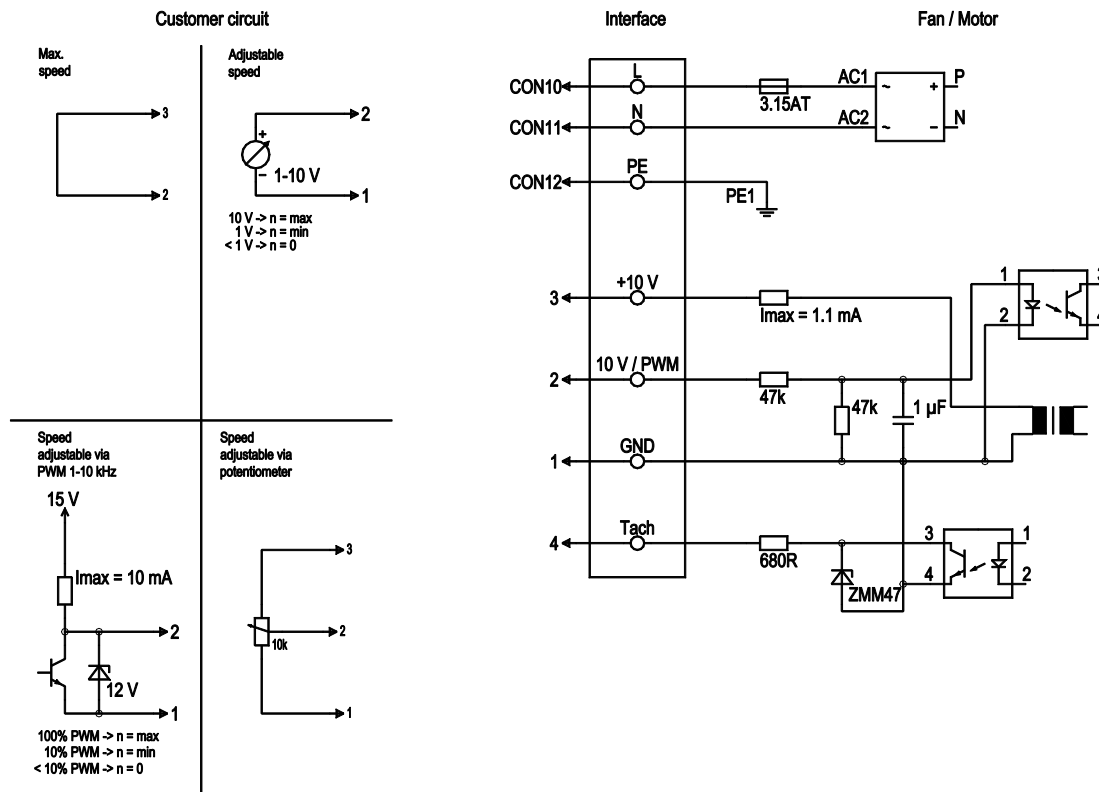
|   |                                                                           |
|---|---------------------------------------------------------------------------|
| 1 | Accessory part: inlet ring 09576-2-4013 not included in scope of delivery |
| 2 | Max. clearance for screw 5 mm                                             |
| 3 | Cable PVC AWG20                                                           |
|   | 2x splice, 1x flat push-on receptacle 6.3x0.8 (PE)                        |
| 4 | Cable PVC AWG22                                                           |
|   | 4x splice                                                                 |

## Accessory part



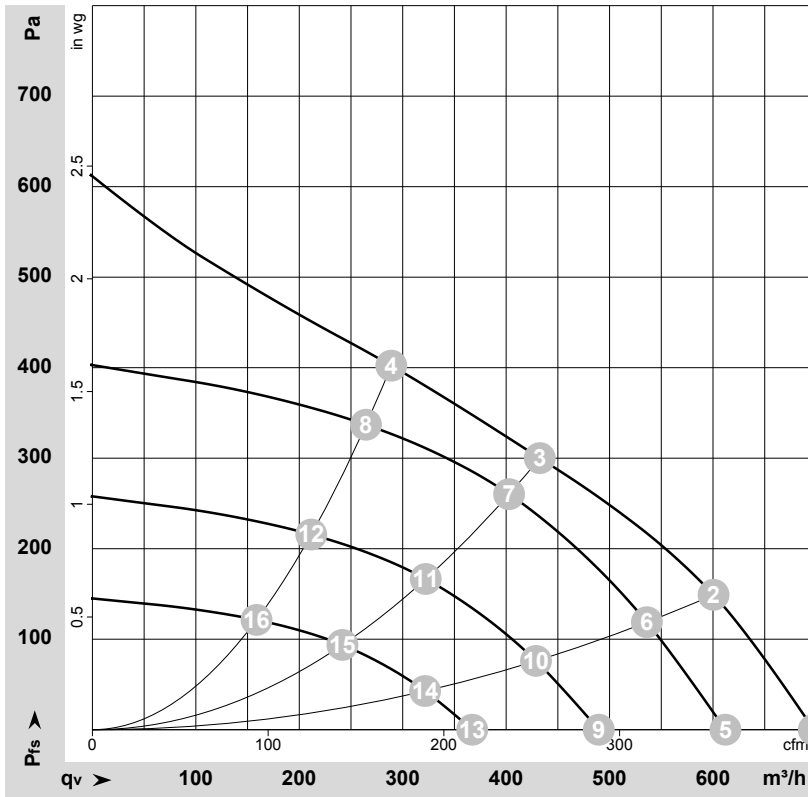
Inlet ring 09576-2-4013

## Connection diagram



| No. | Conn. | Designation        | Color        | Function/assignment                                                        |
|-----|-------|--------------------|--------------|----------------------------------------------------------------------------|
|     | CON10 | L                  | black        | Power supply 230 VAC, 50-60 Hz, see nameplate for voltage range            |
|     | CON11 | N                  | blue         | Neutral conductor                                                          |
|     | CON12 | PE                 | green/yellow | Protective earth                                                           |
|     | 1     | GND                | blue         | GND connection for control interface                                       |
|     | 2     | 0-10V PWM          | yellow       | Control input 0-10 V or PWM, electrically isolated                         |
|     | 3     | 10 V / max. 1,1 mA | red          | Voltage output 10 VDC 1.1 mA, electrically isolated, short-circuit-proof   |
|     | 4     | Tacho              | white        | Tach output: open collector, 1 pulse per revolution, electrically isolated |

## Curves: Air performance 50 Hz



$$\rho = 1.15 \text{ kg/m}^3 \pm 2 \%$$

Measurement: LU-171430-1

Air performance measured according to ISO 5801 installation category A. For detailed information on the measurement setup, contact ebm-papst. Intake sound level: Sound power level according to ISO 13347 / sound pressure level measured at 1 m distance from fan axis. The values given are valid under the specified measuring conditions and may vary due to conditions of installation. For deviations from the standard configuration, the parameters have to be checked on the installed unit.

## Measured values

|    | U   | f  | n                 | P <sub>ed</sub> | I    | LpA <sub>in</sub> | LwA <sub>in</sub> | q <sub>v</sub> | p <sub>fs</sub> | q <sub>v</sub> | p <sub>fs</sub> |
|----|-----|----|-------------------|-----------------|------|-------------------|-------------------|----------------|-----------------|----------------|-----------------|
|    | V   | Hz | min <sup>-1</sup> | W               | A    | dB(A)             | dB(A)             | m³/h           | Pa              | cfm            | in. wg          |
| 1  | 230 | 50 | 3420              | 75              | 0.67 | 66                | 73                | 695            | 0               | 410            | 0.00            |
| 2  | 230 | 50 | 3360              | 79              | 0.70 | 60                | 68                | 600            | 150             | 355            | 0.60            |
| 3  | 230 | 50 | 3200              | 83              | 0.75 | 58                | 66                | 435            | 300             | 255            | 1.20            |
| 4  | 230 | 50 | 3275              | 83              | 0.73 | 60                | 69                | 290            | 400             | 170            | 1.61            |
| 5  | 230 | 50 | 3000              | 51              | 0.45 |                   |                   | 610            | 0               | 360            | 0.00            |
| 6  | 230 | 50 | 3000              | 56              | 0.50 |                   |                   | 535            | 119             | 315            | 0.48            |
| 7  | 230 | 50 | 3000              | 69              | 0.60 |                   |                   | 405            | 261             | 235            | 1.05            |
| 8  | 230 | 50 | 3000              | 64              | 0.56 |                   |                   | 265            | 337             | 155            | 1.35            |
| 9  | 230 | 50 | 2400              | 26              | 0.23 |                   |                   | 490            | 0               | 290            | 0.00            |
| 10 | 230 | 50 | 2400              | 29              | 0.26 |                   |                   | 430            | 76              | 250            | 0.31            |
| 11 | 230 | 50 | 2400              | 35              | 0.31 |                   |                   | 320            | 167             | 190            | 0.67            |
| 12 | 230 | 50 | 2400              | 33              | 0.29 |                   |                   | 210            | 216             | 125            | 0.87            |
| 13 | 230 | 50 | 1800              | 11              | 0.10 |                   |                   | 365            | 0               | 215            | 0.00            |
| 14 | 230 | 50 | 1800              | 12              | 0.11 |                   |                   | 320            | 43              | 190            | 0.17            |
| 15 | 230 | 50 | 1800              | 15              | 0.13 |                   |                   | 240            | 94              | 140            | 0.38            |
| 16 | 230 | 50 | 1800              | 14              | 0.12 |                   |                   | 160            | 121             | 95             | 0.49            |

U = Voltage · f = Frequency · n = Speed (rpm) · P<sub>ed</sub> = Power consumption · I = Current draw · LpA<sub>in</sub> = Sound pressure level intake side · LwA<sub>in</sub> = Sound power level intake side  
 q<sub>v</sub> = Air flow · p<sub>fs</sub> = Pressure increase

